

# Question Paper

Exam Date & Time: 31-Dec-2024 (09:30 AM - 12:30 PM)



## MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.Tech. (COMPUTER SCIENCE & ENGINEERING) EXAMINATIONS - DECEMBER 2024 / JANUARY 2025

SUBJECT: CSE 2121/CSE\_2121 - COMPUTER ORGANIZATION & ARCHITECTURE  
(MAKEUP)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) For the RISC architecture, List 6 addressing mode along with definition and explain the following with an example. (5)  
i) Index addressing mode  
ii) Indirect addressing mode
- 1B) i) List of N numbers are stored in memory. Write an assembly program to multiply list of N and store the product back into the memory, using the RISC instructions. (2)  
ii) Explain the importance of each conditional flag codes during the instruction executions
- 1C) Determine the product of the following numbers by employing the Booth's multiplication algorithm: (3)  
 $M = -9$  and  $Q = 15$
- 2A) Convert the numbers  $A = (86.625)_{10}$  and  $B = (15.125)_{10}$  into its IEEE754 32-bit format and perform the multiplication between them. (3)
- 2B) Apply Restoring Twos Complement Division to divide  $-13$  by  $+3$ . Convert the final answer to its corresponding decimal representation. Show all the necessary steps. (2)
- 2C) Design a Register Transfer Description to find the factorial of a given number using inbus, registers and adder in the Processing Unit. Identify the control points with the corresponding actions taken by the processing unit and draw an efficient state diagram with corresponding control points to solve the given problem. (5)
- 3A) Create the binary listing of the microprogram for the Register Transfer Description given below. (3)  
Show all the necessary steps to arrive at the binary listing.

```
Declare A[8], B[8], C[8], D[8];
      Declare inbus[8];
Start: A ← inbus, C ← inbus, D ← 0;
      B ← inbus;
      B ← B-1;
Loop:  D ← D+A;
      C ← C-1;
      If C > 0 go to Loop;
      A ← D;
      D ← 0;
      B ← B-1;
      If B > 2 go to Loop;
Halt: go to Halt;
```

- 3B) A computer has a small data cache capable of holding eight words and has a byte addressable memory. Each cache block consists of one word and one word comprises of one byte. When a given program is executed, the processor reads data sequentially from the addresses given below (Addresses are given in decimal). Assume that the cache is initially empty. Show the contents of the cache after each memory access, for an associative-mapped cache that uses the LRU replacement algorithm, and compute the hit rate. (5)
- 7200, 7201, 7208, 7213, 7200, 7216, 7208, 7201, 7215, 7201, 7213, 7218, 7220, 7213, 7217, 7219
- 3C) Consider a long sequence of accesses to a disk with an average seek time of 4 ms and an average rotational delay of 2 ms. The average size of a block being accessed is 4K bytes. The data transfer rate from the disk is 34 Mbytes/sec. Assuming that the data blocks are randomly located on the disk, estimate the average percentage of the total time occupied by seek operations and rotational delays. (2)
- 4A) Consider a memory consisting of 64K words of 8 bits each. Organize this memory module using 16K x 1 static memory chips, and calculate, (5)
- i) Number of 16K x 1 static memory chips are required to organize 64K x 8 larger memory.
  - ii) Number of rows and columns are used to organize 64K x 8 larger memory.
  - iii) How many bits are required for a decoder to select rows?
- 4B) A disk unit has 16 recording surfaces. It has a total of 10,000 cylinders. There is an average of 500 sectors per track, and each sector contains 1024 bytes of data. (3)
- i) What is the maximum number of bytes that can be stored in this unit?
  - ii) What is the data transfer rate in bytes per second at a rotational speed of 5400 rpm?
- 4C) How the recent architectures handle uniform memory and non-uniform memory access issues? Differentiate the features. (2)
- 5A) Specify the need of pipelining in modern day processors. List the advantages of using pipelining. Also consider a five-stage pipelined processor (with stages: Instruction Fetch (IF), Instruction Decode (ID), Execute (EX), Memory Access (MEM), and Write Back (WB)), assume that each stage takes 1 cycle to complete. An instruction sequence consists of five independent instructions. How many cycles will it take for the processor to complete all five instructions? Justify your answer with a proper diagram and explanation. (5)
- 5B) A processor supports Fine-Grained Multithreading with a cycle-by-cycle thread switch and has four threads, each with independent instructions. Assume that each instruction takes 5 cycles to complete, and no thread dependencies exist. If each thread begins executing sequentially from the first cycle, how many cycles will it take to complete one instruction from each thread? Provide a detailed explanation for your answer. (3)
- 5C) Consider a multiprocessor system using a cache coherence protocol. If **Processor P1** writes to a cache line, and **Processor P2** has the same cache line in its cache in the **Shared** state, what happens to **Processor P2**'s cache line when **P1** performs the write? Explain the scenario. (2)

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